

sideration of the approach to storage, and also introduce some innovative new solutions that can help protect business, says Ron Riffe, Program Director, IBM Global Storage. All the quotes below are his.

According to the World Economic Forum, from 2008 to 2016, over 20 million people a year have been forced from their homes by extreme weather such as floods, storms, wildfires, and hotter temperatures. These disasters had an impact on business as well. In addition to these disasters, cyberattacks are now one of the top ten long-term risks, high up on the list of both likelihood and impact.

Bad actors are using AI to engineer more potent cyberattacks, and one in four organisations is forecast to experience a data breach in the next two years. A large majority of the World Economic Forum survey respondents expected the risk of cyberattacks to only increase, leading to theft of money and data and breakdowns in information infrastructure.

Interpol (The International Criminal Police Organisation) agrees. It says that in the last year, the target of cybercrime has significantly shifted from individuals to small businesses to major corporations, governments, and critical infrastructure.

Financial Times reported double extortion ransomware attacks—where hackers steal and encrypt on-site sensitive information and threaten to release that data—have increased by 200 percent in 2020. While the median victim size by revenue was \$40 million, more than

a hundred victim organisations had annual revenues in excess of one billion US dollars. Riffe says, cybercrime has become big business.

5 important data resilience questions

With cyber attackers on the loose, data often becomes infected, but the malware can take an average of 197 days (almost six months) to discover.

In order to recover from a situation like this, it is needed that a backup copy is maintained for a long time, such that a clean uninfected copy is always available. Certainly, the first order of business is to prevent bad actors from gaining access to your data in the first place. It's equally important to be able to recover when the inevitable happens.

Commonly, security solutions are designed to stop breaches before they happen. But when the attackers do manage to get through, you want to be prepared to quickly respond. And that gives rise to a number of important questions about data resilience.

1. Is primary storage resilient?

Perhaps the first point to consider here is the place the primary data is stored.

In a traditional recovery scenario, the most recent backup copies are often the ideal ones to restore. That makes it a natural place to start in response to a cyberattack. But remember, cyberattacks present a new wrinkle; the attack that has just been discovered might have happened on an average of six months ago. Hence examining quite a few progressively

older backup copies to find the most recent clean uninfected copy can take extensive work.

Ideally, an organisation can maintain snapshot copies on primary storage for near instant recovery, such that a quick access to the inventory is given for instant mounting of inspection. However, organisations do not maintain months of snapshots on their primary storage. "There are also copies maintained on the secondary storage and you want the same kind of inventory and instant mount capability there, making the job of finding the latest clean copy easier."

2. Is the data copy isolated?

Proper isolation of a backup copy is a major factor that can help retrieve and properly protect the data. When attackers gain access to primary data, often by infecting applications, it is better to keep a copy of the data isolated. It will be beneficial to ensure that an air gap is always there between the application data and its copy. The attacker cannot cross this gap, which keeps copies uninfected.

3. Is the data copy immutable?

If a sophisticated attacker manages to gain access to uninfected copies, a second level of protection can be used to store that data in an unchangeable form. Write Once and Read Many (WORM) technologies can help here.

4. Is the copy discoverable? "If you learn that you are a victim of an attack, it is important to react quickly." In order to do this, a copy of the data should always have been taken before the attacker infects it.

Attackers often penetrate systems and stay. So, when a user realises

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